

A Packing of 13 Four-Spaces in $\mathbb{R}^{11}$

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Abstract

We give 13 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 2.757240. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.740954, obtained by deleting 1 of the 14 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 13 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.757240 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 13$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.740954, obtained by deleting 1 of the 14 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 13/12 = 2.757576$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N13.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
275724044391094866645664175796668494547990813297981924568647689570031996
663346879275943400775576219861480224250971571317086773960129100686838357
3093826900461503384483244081118687399853150574797
```

denominator:

```
999999999999062618495706471755378447945570307325647955640839900127610468
300449808792622550962492364081373501289072148481216962294747353874725538
672412795705541139288488067959685505469747131731
```

B Coordinates

13 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N13.txt`.

```
-518827716276 -45226930123 337127520732 -728734937824477317993 16531748654 -52513678737 8236046212 361962095651 320309611953 212470773753 2133567790
187054925678 300561036583 47962905415 218409107054 406890428922 50958602172 -4799906327 -327174370561 148728100828 169580150061 -320393015817 -40709067550
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